

Inconsistencies in the role of the tongue root in phonological contrasts

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Central questions

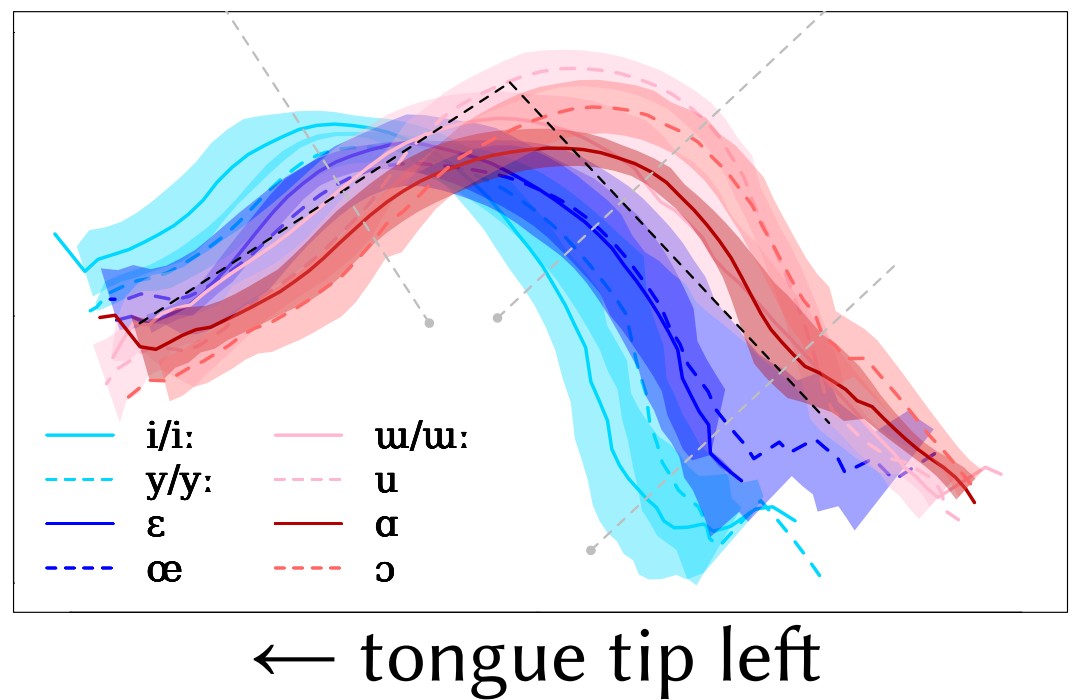
- Is the role of the tongue root (TR) quantifiably consistent with previous studies?

Approach

- Develop quantifiable measure of contribution of various areas of tongue
- Test on various contrasts reported to use TR

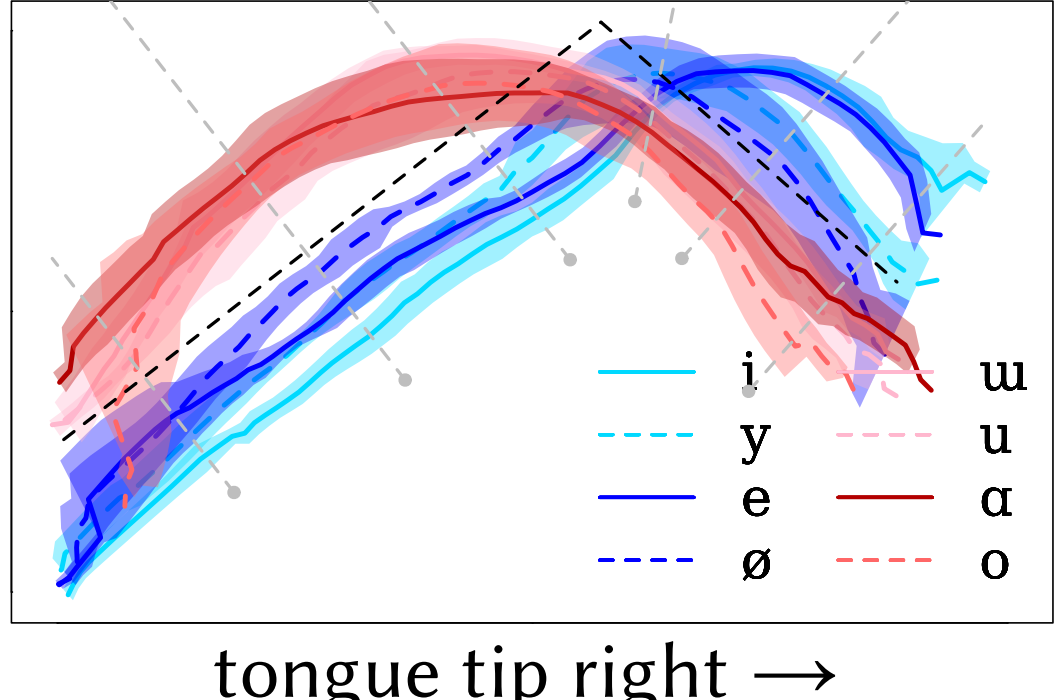
Dolgan

V height: TR↔, TB↑
V backness: TR↔, TB↔
(Banach et al., to appear)



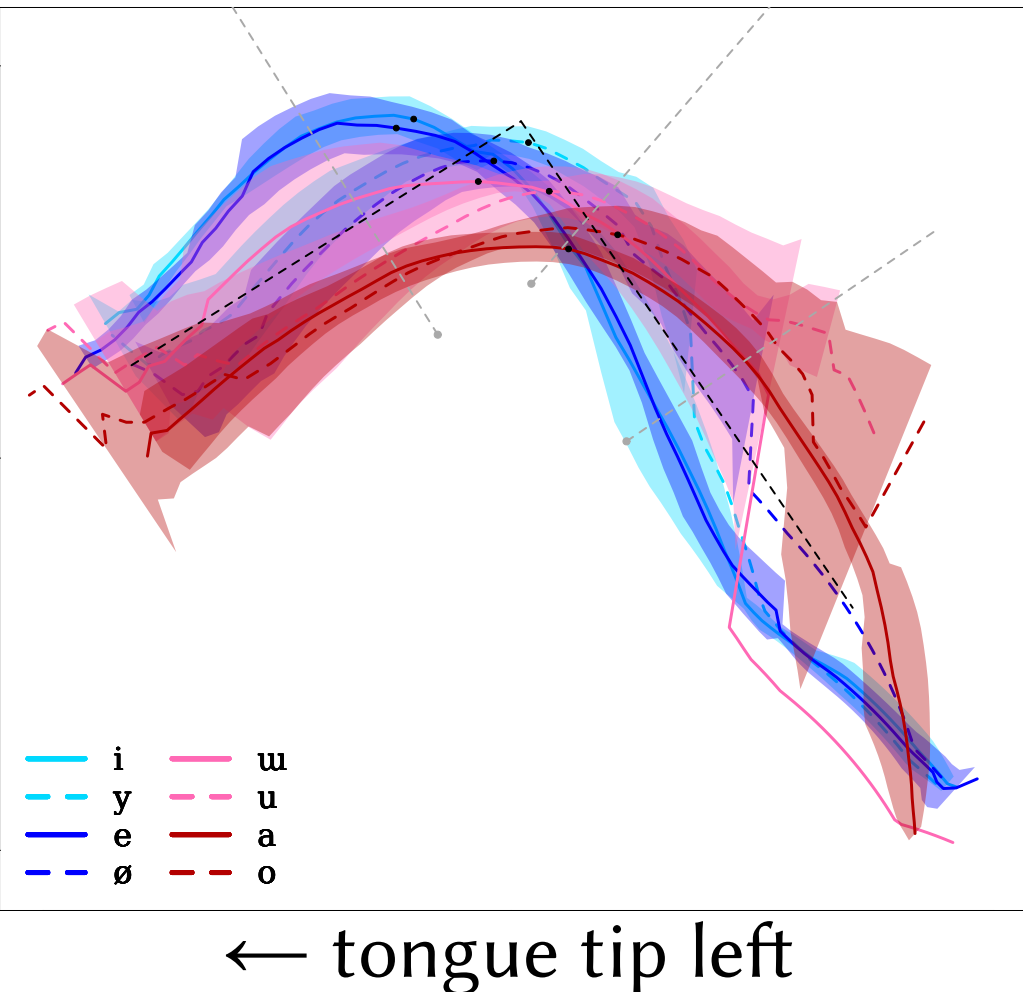
Kyrgyz

V height: TB↑
V backness: TB↔, TR↔
(Washington, 2016, 2019)



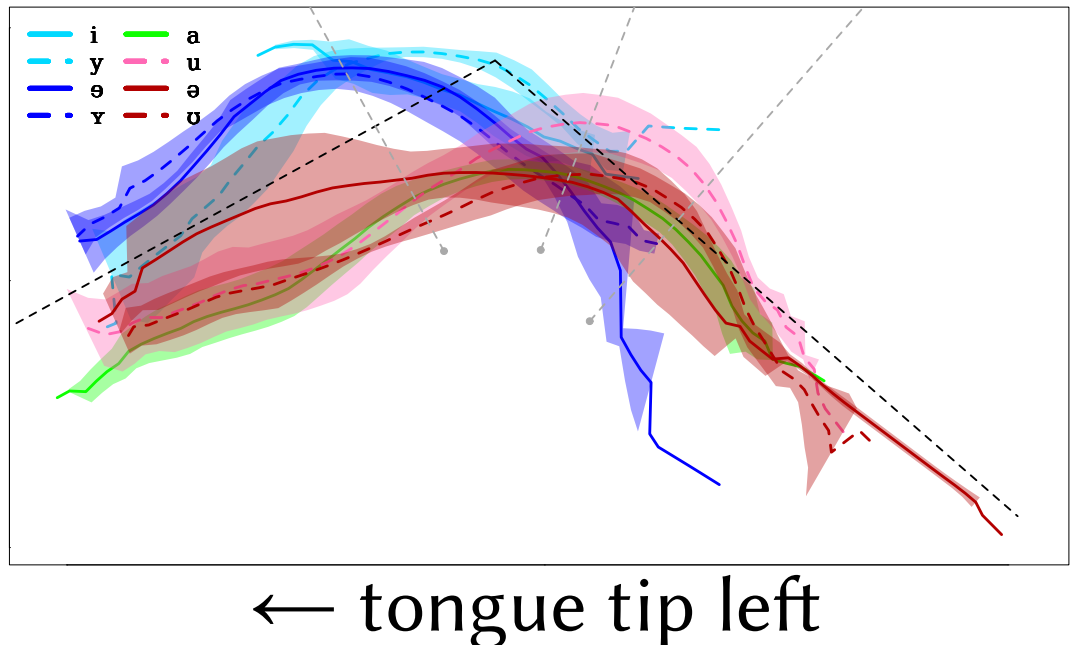
Qumuq

V height: TB↑
V backness: TB↔
(Washington et al., 2020)



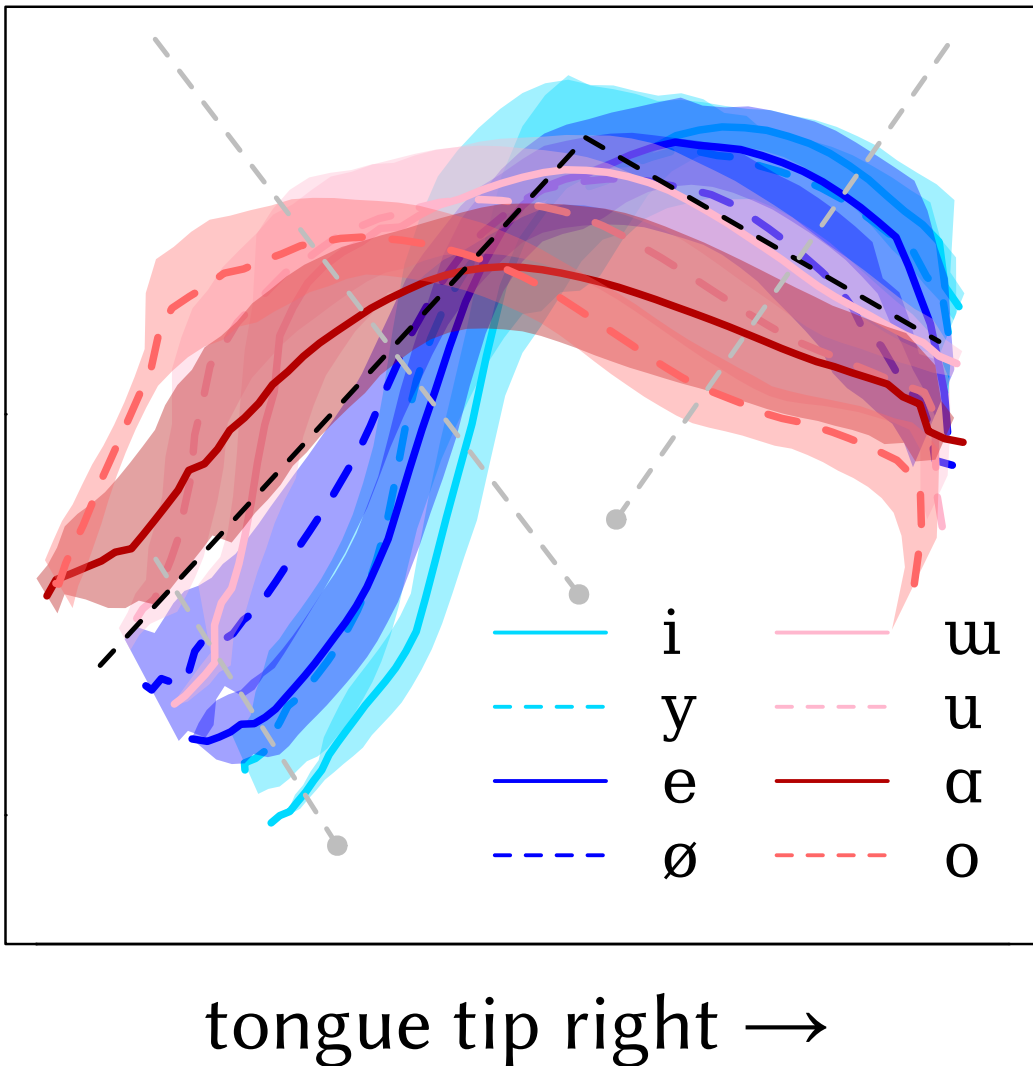
Tatar

V height: TB↑
V backness: TB↔, TR↔
(Washington et al., 2020)



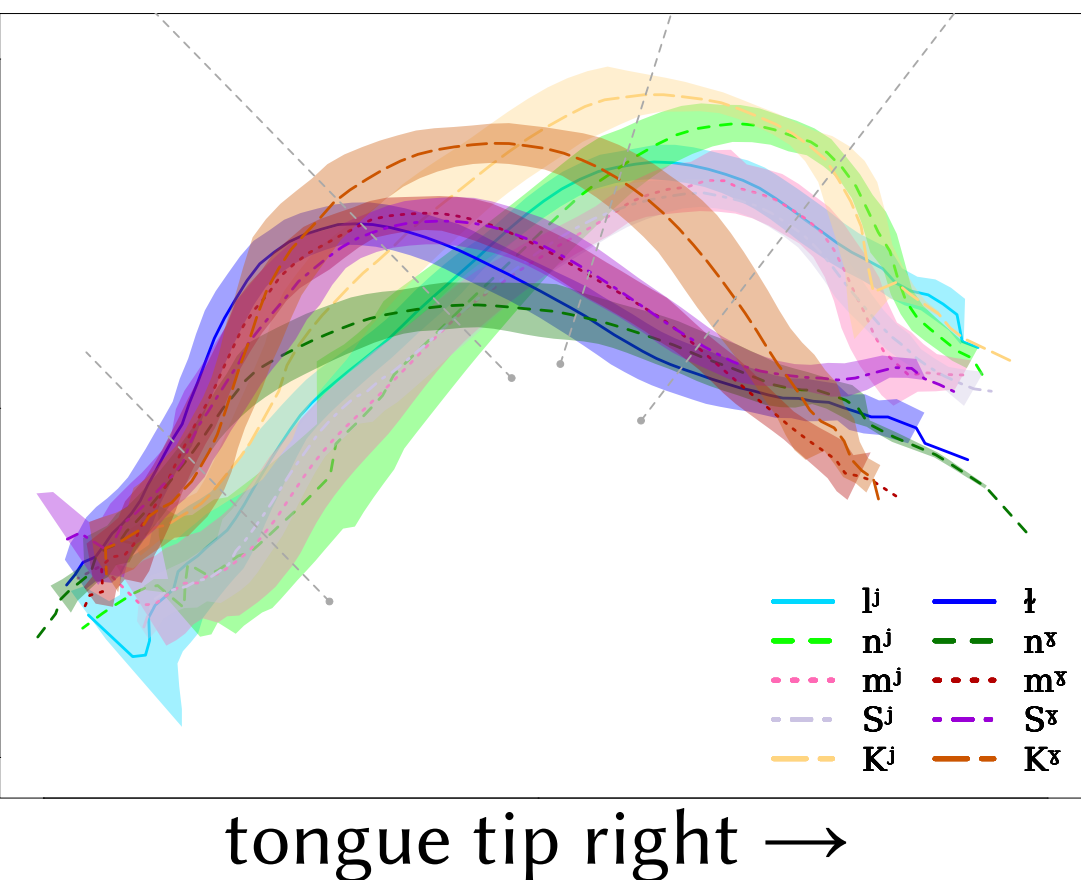
Turkish

V height: TB↑
V backness: TB↔
(Washington, 2016, 2019)



Russian

C^y/C^j: TB↑, TR↔
(Banach et al., 2026, etc.)



Methodology

Data from previous studies (cited)

- Wordlist tasks, in carrier phrases

Annotated midsagittal tongue surface using UltraTrace
(Murphy et al., 2020)

- frame closest to midpoint of Vs

Plotted and **analysed** using ultrapolaRplot (Outkin & Washington, 2024)

- Perpendicular rays intersect various areas of the tongue (mid TB, upper TR, etc.)
- Pairwise t-test between categories at intersections (adjusted for Type I error)
- Pairwise t-test compares maximum curvature (PMC) groups by calculating differences in distance to origin
- PMC comparison is a measure of difference in TB position

Discussion

- TR↔ significant factor in Russian C^y/C^j; does not hold for all pairs
- TR↔ significant factor in all Turkic V backness, but not for each individual pair in each language
- Some areas of TR significant for given contrast, while other areas not
- TR↔ significant for some rounding pairs, likely used to amplify perceptual distinctiveness beyond lip rounding

Future work

- More robust datasets
- Multiple speakers of the same language
- Extent of co-production vs TR independence
- Different analytical methods?

References

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